Resilient Supply Chains

A Graph-Based Optimization Approach

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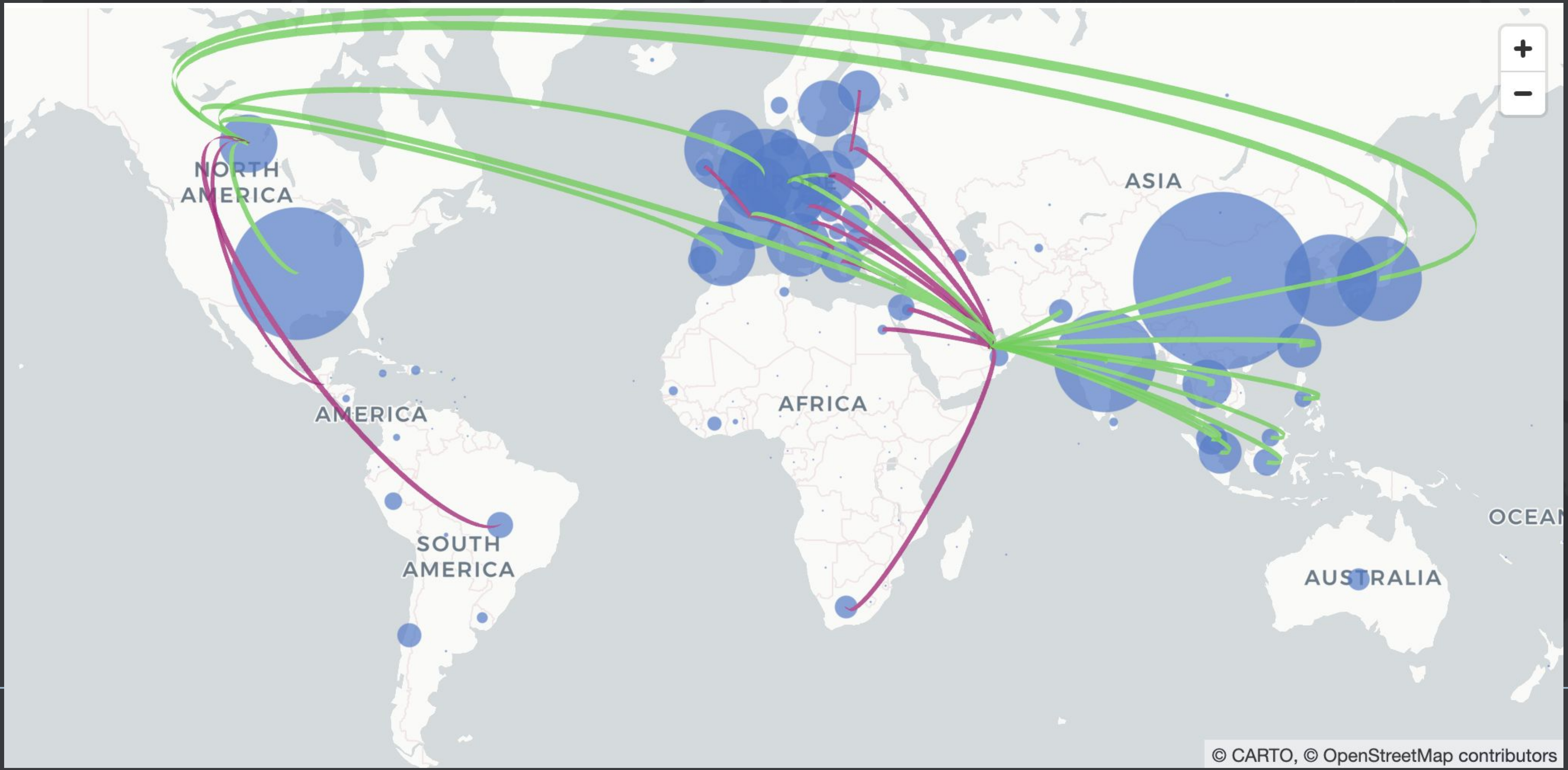
The Problem

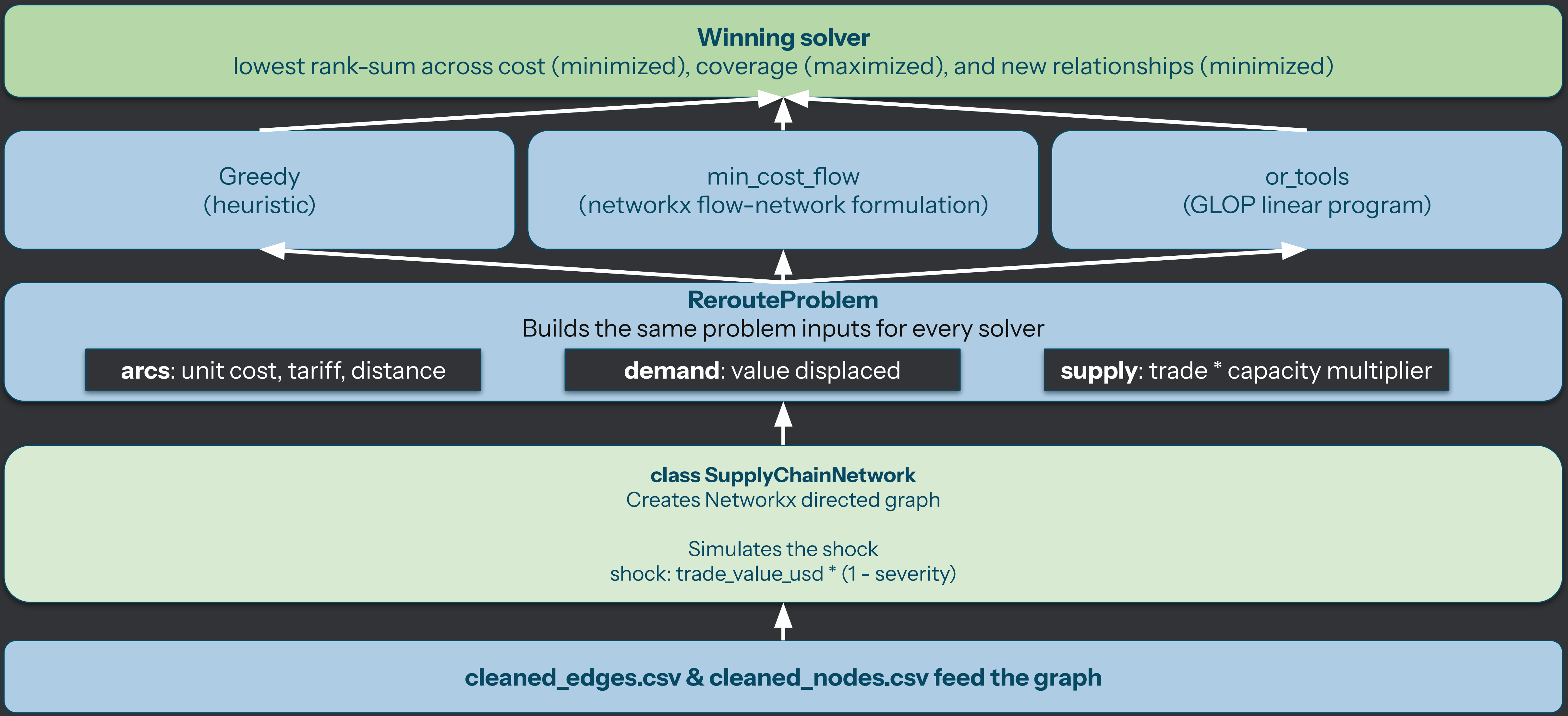
Supply chains are optimized for low costs, not resilience, and lack the flexibility to respond to global shocks, leaving companies exposed when suppliers fail.

The Solution

An interactive directed trade-value graph based on UN Comtrade data that simulates disruptions to compares how different optimization strategies reroute around the shock.



The Architecture



Access the tool here: https://supply-chain-mapping-hys.streamlit.app/

My Goal
Build a portfolio project to learn about supply chains and optimization through graph-based simulations.

Challenges

- Tariff and lead-time data are placeholders formulated with rule-based logic.
- Distances are based on “great-circle” calculations as a placeholder for actual shipping routes and modes of transport.
- Capacity multiplier assumes the same extra capacity across all remaining suppliers, lack of data on actual capacity.
- UN Comtrade represents historic reporting, not live data.

Future Solutions

- Agentic feature for live data pulls, news feeds, and tariff updates.
- Calculate port-level shares within countries for more granular analysis.
- Port-level data enables more accurate shipping estimates/modes of transport.
- Subject matter expert feedback to inform actual costs for onboarding new suppliers.

Thank you

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